
Rotation-Invariant Vision Representations via Graph Neural Networks

Abstract

Modern vision models, though highly accurate, are often brittle under distribution shifts such as image rotations. In real-world applications—e.g., aerial or robotic imagery—objects may appear at arbitrary orientations, revealing this vulnerability. This paper introduces **RotInvNet**, a rotation-invariant representation learning framework that explicitly models spatial relationships between image patches using graph neural networks. Built atop frozen pretrained backbones, RotInvNet constructs a hybrid graph that connects patch tokens through both spatial grid and feature-based k -NN edges. A dedicated GNN module propagates information across this structure and produces robust embeddings, trained via a combination of contrastive and consistency objectives. RotInvNet is effective across multiple vision backbones and yields significant performance gains under diverse rotation settings on both MiniImageNet and CIFAR-10.

1 Introduction

Recent vision models have achieved superior performance in various tasks, such as image classification [1, 2], object detection [3], and image segmentation [4]. However, they may suffer from distribution shifts and tend to fall short in out-of-distribution scenarios, especially when the orientation of objects deviates from the training distribution. While standard datasets like ImageNet contain mostly upright images, real-world scenarios, such as satellite imagery [5] or robotic vision [6], routinely involve objects at arbitrary angles.

A common approach to mitigate this sensitivity is rotation augmentation during training. While such data augmentation can improve robustness, it introduces downsides: increased training cost, potential degradation on canonical (upright) inputs, and limited generalization to unseen rotations or categories. Additionally, mainstream architectures such as ResNet [1] and ViT [2] lack inductive biases for rotation equivariance or invariance.

In this paper, we propose **RotInvNet**, a general-purpose module for learning rotation-invariant visual representations—without modifying or fine-tuning the vision backbone. Our key insight is that object structure, captured by the relative positions and feature similarities among image patches, remains largely stable under rotation. RotInvNet constructs a hybrid graph over patch tokens by combining spatial adjacency and feature-based k -nearest neighbor connectivity. A lightweight graph neural network (GNN) then propagates information across this structure to produce rotation-robust embeddings. Our method is modular, compatible with both convolutional (e.g., ResNet) and transformer-based (e.g., ViT) backbones, and is trained with a combination of contrastive and consistency objectives to explicitly promote invariance.

We evaluate RotInvNet on MiniImageNet and CIFAR-10 using both convolutional (ResNet-50/18) and transformer (ViT-B/16) backbones. Our method consistently improves accuracy and reduces performance variance across rotated test sets—even generalizing to unseen rotated categories. Compared to models trained with rotation augmentation, RotInvNet achieves comparable or better results. These

findings demonstrate that graph-based reasoning over patch relationships leads to rotation-invariant visual representations.

Our main contributions are summarized as follows:

- We propose **RotInvNet**, a modular GNN-based framework for learning rotation-invariant representations, which operates on top of frozen pretrained vision backbones.
- We introduce a hybrid graph construction method that captures both spatial locality and feature-level similarity between image patches.
- We design a multi-objective training scheme combining contrastive, cross-entropy, and consistency losses to explicitly enforce rotation invariance.
- We demonstrate the effectiveness of RotInvNet across multiple backbones, resolutions, and settings, consistently improving rotation robustness with competitive performance.

2 Related Work

Rotation Invariance/Equivariance in General Literature: Rotation invariance and equivariance has appeared throughout many areas of literature. Geared for physics problems, [7] proposes rotating - or rather “unrotating” - all data into a standard configuration. It performs this using a contraction and diagonalization. The now famous work [8] proposes several algorithms for achieving rotation invariance, notably including the use of graph based data representations. The use of differing coordinates systems – especially hyperspherical – have been shown to be effective throughout a variety of applications [9, 10, 11]. While these do not guarantee invariance or equivariance, they do naturally represent the concepts more easily.

Rotation Data Augmentation in CV: A common strategy in literature to address rotation invariance is to train on random rotations of the images in the dataset [12, 13, 14]. Data augmentation alone has been shown to be effective in [14] for handling a variety of invariances. Some works, such as [15] research how to best augment the data based on statistical analyses. However, training times and necessary network size are shown to grow as the invariances are learned. Addressing the relative size of bounding boxes using the augmentation framework is [12], which shows that considering alignment with an ellipse leads to improved box size.

Architectural Modifications in CV: Some works have considered keeping the dataset the same, but instead focus on changing the neural network architecture to handle rotation invariance. An algorithm that uses Gaussian Scale Mixtures to create rotation invariant priors is proposed in [16]. RODFormer [17] proposes identifying rotation in bounding boxes for images by using a rotation-aware loss. This does not create rotation invariance, but rather learns to predict the rotations present in the data. Following a substantially different approach, [18] proposes using B-splines defined on a Lie algebra¹ Through the properties of the algebra, invariance to transforms – like rotations – can be encoded. Other works have trained to have alignment invariant operations, but these work over a discrete partition of the possible rotation space. Example works in this area include [19, 20]. A limitation of this approach is that it is psuedo-invariance; it only captures invariance within the bound of accuracy stemming from the size of the rotation space partitions. To remove the reliance on hand-crafted features, [21] creates learnable filters for rotations; these filters are made through combining radial and sin functions. Once again, this work does not handle all possible rotations.

3 Method

Real-world images are commonly subject to rotation variations, posing substantial challenges for modern computer vision models trained predominantly on upright images. We denote each input image by $x \in \mathbb{R}^{3 \times H \times W}$. Our goal is to learn a function

$$f_\theta : \mathbb{R}^{3 \times H \times W} \rightarrow \mathbb{R}^D \quad (1)$$

such that for any rotation $r \in \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$,

$$z = f_\theta(x), \quad z_r = f_\theta(\text{Rotate}(x, r)), \quad |z - z_r|_2 \approx 0. \quad (2)$$

¹Lie algebras represent symmetry groups.

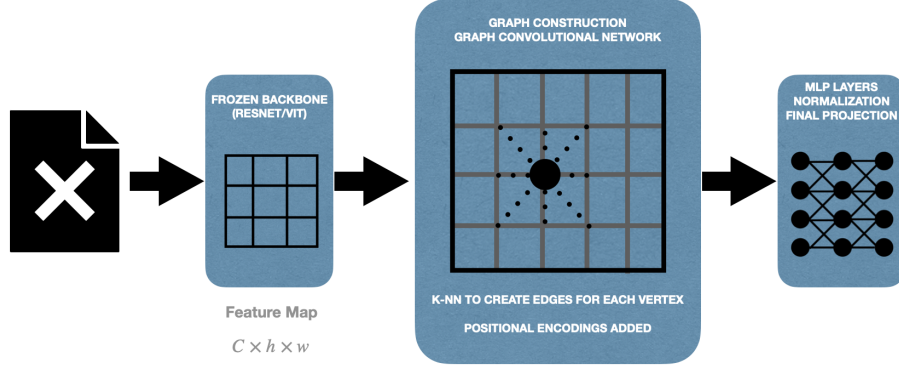


Figure 1: Overarching model architecture

To bridge the performance gap arising from rotational distribution shifts, we propose *RotInvNet*, a rotation-invariant representation learning framework. RotInvNet produces embeddings that are robust to rotations by explicitly modeling the relational structure among visual features using graph neural networks (GNNs). Our method comprises three synergistic components: (i) a hybrid spatial–feature graph construction module, (ii) a GNN-based encoder called *PatchGraphNet*, and (iii) carefully designed joint contrastive and supervised training objectives.

3.1 Hybrid Spatial–Feature Graph Construction

Intuitively, the spatial structure and feature-level similarity among image patches remain largely intact under rotations. Motivated by this insight, we represent each image as a structured graph built upon deep visual features. Specifically, given an input image x , we first extract a spatial feature map using a pretrained, frozen vision foundation model:

$$F = \text{Backbone}(x) \in \mathbb{R}^{C \times h \times w}. \quad (3)$$

Each spatial location of the feature map corresponds to a distinct node in our graph. Formally, we flatten the feature map into $N = h \times w$ patch tokens,

$$\{x_i\}_{i=1}^N, \quad x_i \in \mathbb{R}^C, \quad (4)$$

each positioned at grid coordinates (u, v) .

To comprehensively capture local and global relational structures, we construct two complementary sets of edges:

- **Spatial grid edges** (E_{grid}): Nodes are connected to their immediate neighbors $(u \pm 1, v)$ and $(u, v \pm 1)$, preserving local spatial context.

$$E_{\text{grid}} = \{(i, j) \mid |u_i - u_j| + |v_i - v_j| = 1\}. \quad (5)$$

- **Feature k-nearest neighbor (kNN) edges** (E_{knn}): Each node is further connected to its top- k most similar neighbors based on cosine similarity:

$$E_{\text{knn}} = \{(i, j) \mid j \in \text{TopK}(\cos(x_i, x_j), k)\}. \quad (6)$$

Moreover, we incorporate explicit positional encodings $p_i \in [-1, 1]^2$ for each node to inform the network of absolute spatial positions:

$$p_i = \left(\frac{2u}{h-1} - 1, \frac{2v}{w-1} - 1 \right). \quad (7)$$

The resulting hybrid graph combines spatial regularity and feature-driven connectivity:

$$G = (V, E), \quad V = \{1, \dots, N\}, \quad E = E_{\text{grid}} \cup E_{\text{knn}}. \quad (8)$$

3.2 PatchGraphNet: Graph-based Encoding Module

Given our constructed graph, we encode node representations through the *PatchGraphNet*, a multi-layer GNN designed to aggregate information across neighboring patches. Each node’s initial embedding combines its visual and positional information:

$$h_i^{(0)} = [x_i; p_i] \in \mathbb{R}^{C+2}. \quad (9)$$

We propagate embeddings through L graph convolutional layers:

$$h_i^{(\ell+1)} = \sigma \left(W^{(\ell)} \sum_{j \in \mathcal{N}(i)} h_j^{(\ell)} + b^{(\ell)} \right), \quad (10)$$

where $\mathcal{N}(i)$ denotes neighbors of node i , and $\sigma = \text{ReLU}$. After the final layer, we pool the node embeddings into a single representation capturing global image structure:

$$z = \frac{1}{N} \sum_{i=1}^N h_i^{(L)} \in \mathbb{R}^d. \quad (11)$$

To generate the final rotation-invariant embedding, we apply a two-layer projection head and normalization:

$$z' = W_2 (\text{ReLU}(W_1 z)), \quad \hat{z} = \frac{z'}{\|z'\|_2} \in \mathbb{R}^D. \quad (12)$$

3.3 Joint Learning Objectives

To explicitly encourage rotation invariance and robust visual discrimination, we introduce a set of coordinated training objectives. During training, each mini-batch image x_i is augmented by generating rotated views $\{x_i^v = \text{Rotate}(x_i, r_v)\}_{v=1}^V$, with distinct rotations sampled from the set $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$. Denote the embeddings for these views as $\hat{z}_i^v = f_\theta(x_i^v)$. We propose three losses to enhance the rotation-invariance while maintaining the discriminative capability.

Supervised Contrastive Loss. Inspired by [22], we adopt a supervised contrastive loss to cluster representations of the same class regardless of rotation, thereby learning robust semantic groupings:

$$\mathcal{L}_{\text{sup}} = - \sum_{i=1}^B \sum_{v=1}^V \frac{1}{|P(i, v)|} \sum_{(p, u) \in P(i, v)} \log \frac{\exp(\hat{z}_i^v \cdot \hat{z}_p^u / \tau)}{\sum_{(a, m) \neq (i, v)} \exp(\hat{z}_i^v \cdot \hat{z}_a^m / \tau)}, \quad (13)$$

where $P(i, v) = \{(j, u) \mid y_j = y_i, (j, u) \neq (i, v)\}$.

Rotation Consistency Loss. The same image with different views should have similar embeddings. We explicitly penalize divergence between the embeddings of the original and rotated images:

$$\mathcal{L}_2 = \frac{1}{B} \sum_{i=1}^B \sum_{v=2}^V \|\hat{z}_i^1 - \hat{z}_i^v\|_2^2. \quad (14)$$

Cross-Entropy Loss. To retain the inherent performance of the backbone model, a supervised cross-entropy loss further guides class-discriminative feature learning on unrotated images through a temporary classification head during training:

$$\mathcal{L}_{\text{ce}} = - \sum_{i=1}^B \log [\text{softmax}(W_c \hat{z}_i^1)]_{y_i}. \quad (15)$$

The overall objective balances semantic alignment, discriminative capability, and rotation invariance:

$$\boxed{\mathcal{L} = \mathcal{L}_{\text{sup}} + \lambda_{\text{ce}} \mathcal{L}_{\text{ce}} + \lambda_2 \mathcal{L}_2.} \quad (16)$$

Table 1: Top-1 accuracy (%) on MiniImageNet across different rotation angles. All models are evaluated using linear probing including our RotInvNet. “Train” indicates the angle used during representation learning. We report mean (Avg) and standard deviation (SD) across 4 test rotations.

Model	Train	Validation						Test					
		0°	90°	180°	270°	Avg	SD	0°	90°	180°	270°	Avg	SD
RN50	0	95.72	74.72	77.64	74.84	80.73	10.08	91.98	73.22	75.10	75.30	78.90	8.77
Ours	0	92.93	82.62	83.23	83.26	85.51	4.96	89.44	81.76	81.44	81.92	83.64	3.87
RN50	90	92.60	84.66	79.43	84.50	85.30	5.44	88.76	82.68	76.34	82.38	82.54	5.07
Ours	90	92.92	82.89	83.28	83.49	85.65	4.86	89.46	81.84	81.46	81.84	83.65	3.88
RN50	180	93.81	77.87	86.37	77.67	83.93	7.73	90.12	75.20	83.94	75.60	81.22	7.17
Ours	180	92.88	82.88	83.36	83.41	85.63	4.84	89.54	82.02	81.44	81.90	83.73	3.88
RN50	270	93.06	84.26	79.96	84.38	85.42	5.50	88.94	82.70	77.32	82.70	82.92	4.75
Ours	270	92.89	83.01	83.38	83.48	85.69	4.80	89.38	81.84	81.54	81.82	83.65	3.83

4 Experimental Setup

Datasets. We evaluate our proposed method on two widely-used benchmark datasets. **MiniImageNet** [23] is a popular subset of ImageNet, consisting of 100 classes. The images are resized to 224×224 pixels. The standard dataset splits include 500 images per class for training, 100 images per class for validation, and 50 images per class for testing. **CIFAR-10** [24] comprises 60,000 color images distributed across 10 classes (50,000 for training, 10,000 for testing). Each image is 32×32 pixels. To explicitly evaluate rotation robustness, we augment both datasets by generating rotated variants of images at discrete angles: $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$.

Backbone Architectures. We use three distinct backbone models to demonstrate the versatility of our approach. **ResNet-50** [1]: A 50-layer convolutional network pretrained on ImageNet-1K, serving as our primary backbone for experiments on MiniImageNet. **Vision Transformer (ViT-B/16)** [2]: A transformer-based vision model pretrained on ImageNet-1K, validating the effectiveness of our approach beyond convolutional architectures. **ResNet-18** [1]: A compact residual network pretrained on ImageNet-1K, specifically adapted for CIFAR-10’s smaller image size. In all experiments, backbone weights remain frozen during training, with only the GNN module and projection head (discarded at inference) parameters updated.

Evaluation Protocol. We adopt the linear probing evaluation protocol to measure the representational quality of learned embeddings. Specifically, after training RotInvNet, we freeze the entire feature extractor (including the GNN module) and append a linear classification head. The linear classifier is trained using cross-entropy loss on the training set. We report top-1 classification accuracy on each rotated test set ($0^\circ, 90^\circ, 180^\circ, 270^\circ$), quantifying both discriminative power and robustness to rotations.

Implementation Details. During training, we use the AdamW optimizer with learning rate 1×10^{-3} , betas (0.9, 0.999), and weight decay 1×10^{-4} . Training lasts for 50 epochs with batch sizes of 512 for MiniImageNet and CIFAR-10. The augmentation pipeline applied during training includes random horizontal flipping, color jittering, Gaussian blurring, resizing, and center cropping. At evaluation, augmentations are simplified to resizing and center cropping only.

5 Experimental Results

This section evaluates the effectiveness of our proposed RotInvNet from multiple perspectives. We first demonstrate its robustness by assessing performance on rotated test sets. We then explore its ability to generalize to previously unseen, out-of-distribution rotated categories. Lastly, we validate the versatility of our method by extending evaluations to different architectures and smaller image resolutions.

Table 2: Top-1 accuracy (%) comparison with random rotation fine-tuning. “LP” refers to the Linear Probe baseline trained only with 0° images. “RR” denotes random rotation augmentation.

Model	Ep.		Validation						Test					
			0°	90°	180°	270°	Avg	SD	0°	90°	180°	270°	Avg	SD
LP	RN50	/	95.72	74.72	77.64	74.84	80.73	10.08	91.98	73.22	75.10	75.30	78.90	8.77
RR	RN50	10	94.98	84.04	84.95	83.91	86.97	5.36	91.34	82.38	81.94	82.16	84.46	4.59
RR	RN50	20	94.95	84.65	84.56	84.16	87.08	5.25	91.12	82.28	82.22	81.96	84.40	4.49
RR	RN50	30	94.78	84.66	85.12	84.40	87.24	5.04	90.36	82.46	82.14	82.12	84.27	4.06
RR	RN50	40	94.74	84.83	84.92	84.38	87.22	5.02	90.94	82.88	82.80	82.46	84.77	4.12
LP	Ours	/	92.93	82.62	83.23	83.26	85.51	4.96	89.44	81.76	81.44	81.92	83.64	3.87
RR	Ours	20	93.34	83.85	84.18	83.65	86.26	4.73	89.54	82.08	81.74	81.68	83.76	3.86

5.1 Classification on the Rotated Images

Table 1 reports the top-1 classification accuracy on MiniImageNet under various training and testing rotation configurations. For each experiment, we perform linear probing using either a pretrained ResNet-50 or our proposed RotInvNet, training on a fixed rotation angle and evaluating across all rotated validation and test sets. When using ResNet-50, training on a specific orientation slightly improves accuracy on that rotation (e.g., training on 90° improves 90° test performance), but this can not help the performance on other orientations, including the canonical 0° view. Moreover, the accuracy variance across rotations is relatively high, indicating that the representations are sensitive to input orientation. In contrast, RotInvNet consistently achieves higher average accuracy across all rotated views and exhibits lower standard deviation. Notably, it maintains strong performance on the original 0° split, while substantially improving results on rotated inputs. It proves that RotInvNet learns rotation-robust representations without sacrificing performance on standard, upright images.

A straightforward approach to mitigate rotation sensitivity is to apply random rotation augmentation during training. As shown in Table 2, this strategy improves the performance of a pretrained ResNet-50 across all rotated validation and test sets. Notably, performance improves consistently up to 20 epochs but begins to saturate beyond that point, indicating diminishing returns with longer training. On the contrary, our RotInvNet achieves comparable average accuracy without any rotation augmentation. Specifically, our method matches the performance of a ResNet-50 trained with rotation augmentation while yielding lower standard deviation, suggesting more stable and rotation-consistent representations (see second-last row). Interestingly, applying random rotation augmentation to our RotInvNet provides little to no additional benefit (see last row). This implies that RotInvNet has already internalized rotation invariance through its architecture and training objectives, rendering additional augmentation redundant. We hypothesize that the residual performance ceiling may stem from two limitations: (i) the inherent rotation sensitivity of the frozen ImageNet-pretrained backbone, and (ii) the linear probing setup used during evaluation (*i.e.*, the linear layer), which lacks the capacity to fully adapt to residual spatial misalignment.

5.2 Generalization to Unrotated Categories

Table 3 illustrates the effectiveness of our rotation-invariant representations on out-of-distribution categories. To assess whether the model truly learns rotation-invariant structure rather than simply memorizing rotated samples, we construct a partially augmented training set in which only a subset of categories is exposed to rotation. When trained on this setup, the baseline ResNet-50 exhibits a substantial accuracy gap between augmented and non-augmented categories. This suggests that its robustness stems from memorizing the specific rotated instances seen during training, rather than acquiring an abstract understanding of spatial structure. In contrast, RotInvNet achieves significantly more balanced performance across both augmented and unaugmented categories. Even without direct exposure to rotated examples from some categories, it generalizes well to their rotated test instances. This supports our claim that RotInvNet learns underlying spatial relationships between image patches, rather than simply encoding rotation-specific visual cues, thus enabling generalization to unseen rotation patterns and novel classes.

Table 3: Top-1 accuracy (%) on MiniImageNet for category-wise rotation robustness. “LP” represents linear probe. “RCR” stands for Random Category Rotation Augmentation, which involves training with rotation applied to a random proportion of categories. We report performance on all validation/test splits, as well as on augmented and non-augmented categories separately.

	Model	Aug.	Validation						Val Aug Categories						Val Non-Aug Categories					
			0°	90°	180°	270°	Avg	SD	0°	90°	180°	270°	Avg	SD	0°	90°	180°	270°	Avg	SD
LP	RN50	/	95.72	74.72	77.64	74.84	80.73	10.08	-	-	-	-	-	-	95.49	66.29	71.39	66.29	74.87	13.96
RCR	RN50	0.25	95.22	72.53	76.03	72.47	79.06	10.90	94.40	91.24	89.96	91.00	91.65	1.92	95.30	63.50	68.46	62.74	72.50	15.41
RCR	RN50	0.50	94.94	76.57	78.76	76.24	81.63	8.95	94.58	89.64	89.06	89.74	90.76	2.57	95.08	65.12	69.68	64.52	73.60	14.50
RCR	RN50	0.75	94.94	80.82	81.92	80.47	84.54	6.96	94.89	86.05	86.00	85.79	88.18	4.47	95.08	65.12	69.68	64.52	73.60	14.50
LP	Ours	/	92.93	82.62	83.23	83.26	85.51	4.96	-	-	-	-	-	-	-	-	-	-	-	-
RCR	Ours	0.25	93.24	83.66	83.54	83.26	85.93	4.88	94.60	87.04	86.20	86.32	88.54	4.06	92.79	82.53	82.65	82.24	85.05	5.16
RCR	Ours	0.50	93.25	83.59	83.61	83.30	85.94	4.88	94.40	85.50	85.70	85.48	87.77	4.42	92.10	81.68	81.52	81.12	84.11	5.34
RCR	Ours	0.75	92.23	83.59	83.85	83.37	85.76	4.32	93.57	83.61	84.13	83.31	86.16	4.95	92.20	83.52	83.00	83.56	85.57	4.43
			Test						Test Aug Categories						Test Non-Aug Categories					
	Model	Aug.	0°	90°	180°	270°	Avg	SD	0°	90°	180°	270°	Avg	SD	0°	90°	180°	270°	Avg	SD
LP	RN50	/	91.98	73.22	75.10	75.30	78.90	8.77	-	-	-	-	-	-	-	-	-	-	-	-
RCR	RN50	0.25	91.18	70.26	72.52	70.38	76.09	10.12	89.04	88.56	86.96	89.28	88.46	1.04	91.89	64.16	67.71	64.08	71.96	13.39
RCR	RN50	0.50	90.90	74.74	75.32	73.76	78.68	8.17	90.96	87.68	86.44	87.96	88.26	1.92	90.84	61.80	64.20	59.56	69.10	14.62
RCR	RN50	0.75	90.68	78.52	79.68	78.30	81.80	5.95	90.64	84.19	83.89	83.89	85.65	3.33	90.80	61.52	67.04	61.52	70.22	13.96
LP	Ours	/	89.44	81.76	81.44	81.92	83.64	3.87	-	-	-	-	-	-	-	-	-	-	-	-
RCR	Ours	0.25	89.40	81.56	81.10	81.36	83.36	4.03	91.20	83.68	83.60	82.72	85.30	3.96	88.80	80.85	80.27	80.91	82.71	4.07
RCR	Ours	0.50	89.24	81.48	81.24	81.32	83.32	3.95	90.72	83.60	83.72	83.36	85.35	3.58	87.76	79.36	78.76	79.32	81.30	4.32
RCR	Ours	0.75	89.30	81.64	81.32	81.48	83.44	3.91	90.08	82.24	82.05	82.08	84.11	3.98	86.96	79.84	79.12	79.68	81.40	3.72

Table 4: Top-1 accuracy (%) of ViT-B/16 and RotInvNet under different training rotations on MiniImageNet. Models are trained on a fixed rotation and evaluated across all rotation angles. RotInvNet demonstrates lower variance and more robust generalization.

Model	Train	Validation						Test					
		0°	90°	180°	270°	Avg	SD	0°	90°	180°	270°	Avg	SD
ViT-B/16	0	98.83	82.47	87.15	82.65	87.78	7.68	94.22	77.40	80.96	77.40	82.50	7.99
Ours	0	95.33	86.92	88.35	86.95	89.39	4.02	91.06	85.08	85.02	84.62	86.45	3.08
ViT-B/16	90	98.41	90.55	89.99	90.58	92.38	4.03	93.48	86.98	84.62	86.38	87.87	3.87
Ours	90	95.40	86.98	88.42	87.10	89.48	4.00	91.16	85.12	85.12	84.78	86.55	3.08
ViT-B/16	180	98.56	86.09	92.53	85.80	90.75	6.07	93.62	80.94	87.52	80.58	85.67	6.19
Ours	180	95.33	86.92	88.43	87.12	89.45	3.98	91.10	85.00	84.98	84.70	86.45	3.11
ViT-B/16	270	98.36	90.39	89.96	90.59	92.33	4.03	93.22	86.64	84.52	86.24	87.66	3.82
Ours	270	95.35	87.00	88.37	87.05	89.44	3.99	91.20	85.12	85.10	84.72	86.54	3.12

5.3 Extension to Vision Transformers

To assess the generality of our approach beyond convolutional architectures, we integrate our GNN module with a pretrained Vision Transformer (ViT-B/16) [2]. The transformer is frozen throughout training. Unlike standard usage where only the [CLS] token is used for classification, we treat the [CLS] token as an additional node in the graph, connecting it to all patch tokens. The resulting graph includes: (i) a 2D spatial grid of patch tokens arranged in a 14×14 layout, (ii) a feature-based k -NN graph among patches, and (iii) global edges linking the [CLS] node to every patch. We inject 2D positional encodings into each node (assigning (0, 0) to the [CLS] node) and apply the GNN over the entire graph. The pooled GNN output replaces the original [CLS] token in downstream prediction.

Table 4 presents top-1 accuracy across rotation settings for both standard ViT-B/16 and our GNN-enhanced RotInvNet. While our method slightly underperforms the baseline on the original 0° view, it substantially reduces variance across rotations. This lower standard deviation indicates stronger rotation invariance, suggesting that RotInvNet compensates for the transformer’s lack of inductive bias toward spatial consistency.

We attribute the small performance degradation on unrotated inputs to several factors. First, the baseline ViT-B/16 already achieves superior performance, leaving less headroom for improvement and increasing sensitivity to architectural perturbations. Second, the ViT’s lower patch resolution (compared to ResNet feature maps) results in smaller graphs, limiting the GNN’s ability to model fine-grained spatial structure. Finally, ViT representations are typically optimized for [CLS] usage, and extracting useful information from raw patch embeddings may require a larger or more specialized

Table 5: Top-1 accuracy (%) of ResNet-18 and Our RotInvNet trained and evaluated on CIFAR-10 with rotated test sets. RotInvNet improves both average accuracy and stability across all orientations.

Model	Train	0°	90°	180°	270°	Avg	SD
ResNet-18	0	43.78	26.27	32.83	26.81	32.42	8.13
Ours	0	52.90	40.14	43.80	41.50	44.59	5.75
ResNet-18	90	27.84	37.80	26.94	34.84	31.86	5.31
Ours	90	47.36	47.10	42.88	47.14	46.12	2.16
ResNet-18	180	31.76	26.43	41.48	26.40	31.52	7.10
Ours	180	50.82	43.81	45.71	43.77	46.03	3.32
ResNet-18	270	29.88	37.94	30.16	40.49	34.62	5.41
Ours	270	46.94	46.74	43.06	47.01	45.94	1.92

GNN. Despite these challenges, RotInvNet consistently improves rotation robustness with only a minor trade-off in overall performance.

5.4 Adaptation to Low-Resolution Images

Graph-based architectures are often constrained by input resolution, as fewer spatial patches lead to smaller graphs and weaker structural representations. To demonstrate the flexibility of RotInvNet, we extend our method to low-resolution inputs using the CIFAR-10 dataset (32×32 images). Applying a deep backbone such as ResNet-50 results in overly coarse spatial features (e.g., 4×4 feature maps), limiting the size and expressiveness of the graph. Therefore, we replace ResNet-50 with a smaller ResNet-18 backbone and modify its architecture to preserve fine-grained spatial detail. Specifically, we replace the initial 7×7 convolution and max-pooling layers with a 3×3 convolution (stride 1) and eliminate early downsampling. This yields higher-resolution feature maps (e.g., 8×8 patches), allowing the GNN to operate on a sufficiently large and informative graph. All other components of RotInvNet remain unchanged.

As shown in Table 5, our method yields substantial gains over the ResNet-18 baseline across all rotation angles. Unlike the results from deeper models, RotInvNet also improves accuracy on the canonical 0° test set, likely due to the relatively small capacity of ResNet-18: our additional GNN layers significantly expand the model’s expressiveness, enabling the learning of more complex spatial dependencies. More importantly, RotInvNet produces consistently strong results across rotated inputs, achieving both higher average accuracy and significantly lower variance, underscoring its robustness on low-resolution inputs.

6 Conclusion

We present **RotInvNet**, a simple yet effective framework for learning rotation-invariant visual representations through graph neural networks. By modeling both spatial and feature-based relationships among image patches, RotInvNet can be seamlessly integrated with various frozen pretrained backbones. The proposed method achieves improved robustness to rotation while maintaining strong performance on canonical views. Extensive experiments on MiniImageNet and CIFAR-10 validate its effectiveness across diverse architectures, image resolutions, and category splits. Our results demonstrate that explicit modeling of patch-level structure enables generalization beyond data-driven memorization of rotations, offering a lightweight alternative to heavy data augmentation or architectural overhauls.

Limitations and future work. While RotInvNet improves rotation robustness, several limitations remain. First, we evaluate on relatively small-scale datasets; future work may explore scaling to large benchmarks like ImageNet [25]. Second, although effective, the performance gap with strong ViT baselines under canonical views suggests a need for deeper or more expressive GNN modules. Third, our current setting considers only discrete rotations, whereas real-world rotations are often continuous. Extending RotInvNet to arbitrary-angle invariance is a promising direction. Lastly, applying this approach to downstream tasks such as detection or segmentation may further unlock the potential of structure-aware, rotation-robust representations.

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